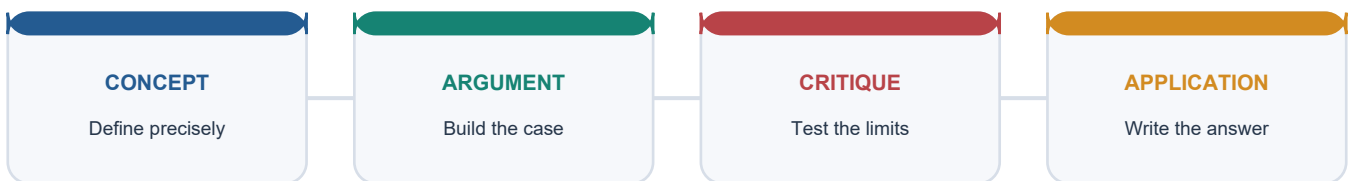


COMPLETE LEARNING SESSION

Rates-Motion-Time-and-Geometry - CSAT Learner-v2 Semantic Successor

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / SESSION INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Rates-Motion-Time-and-Geometry - Complete CSAT Learning Session 5

BASIC LEARNING SESSION 5

Twelve-panel ASCII master flow	5
Canonical Basic owner	7
Rates, Motion, Time and Geometry - MUST-DO	7
1. Visual foundation	7
2. Essential formulas	7
3. Method	8
4. Conditions to respect	9
5. Original solved examples	9
6. Must-Know facts	10
7. Common traps	11
8. Quick checks	11
9. Mini-drill (with answers and explanations)	11
10. Timed transfer, diagnosis and retry gate	12
11. Study links	12
2026 PYQ Integration	12
Recent PYQ Integration (2024-2025)	14
Historical PYQ Integration (2018-2023)	16
Semantic-completeness closure - 2026-09-06	21

BASIC MCQS / REMEDIATION 22

Diagnostic and core set	22
Q1. Practice variant 1: A finishes a job in 12 days and B in 18 days. Working together, about how many days do they take (nearest day)?	22
Q2. Practice variant 1: A pipe fills a tank in 10 hours and a leak empties it in 20 hours. How many hours together (nearest hour)?	22
Q3. Practice variant 1: A vehicle travels at 45 km/h for 4 hours. What distance does it cover?	22
Q4. Practice variant 1: A train 120 m long crosses a pole at 54 km/h. How many seconds does it take?	22
Q5. Practice variant 1: A boat moves at 12 km/h in still water and the stream is 2 km/h. What is its downstream speed?	22
Q6. Practice variant 1: What is the smaller angle between clock hands at 2:20?	22

Q7. Practice variant 1: If today is Monday, what weekday is it after 100 days?	22
Q8. Practice variant 1: What is the area in square metres of a rectangle 12 m by 8 m?	22
Q9. Practice variant 2: A finishes a job in 13 days and B in 19 days. Working together, about how many days do they take (nearest day)?	22
Q10. Practice variant 2: A pipe fills a tank in 11 hours and a leak empties it in 22 hours. How many hours together (nearest hour)?	23
Q11. Practice variant 2: A vehicle travels at 50 km/h for 4 hours. What distance does it cover?	23
Q12. Practice variant 2: A train 130 m long crosses a pole at 54 km/h. How many seconds does it take?	23
Q13. Practice variant 2: A boat moves at 13 km/h in still water and the stream is 2 km/h. What is its downstream speed?	23
Q14. Practice variant 2: What is the smaller angle between clock hands at 3:20?	23
Q15. Practice variant 2: If today is Monday, what weekday is it after 101 days?	23
Q16. Practice variant 2: What is the area in square metres of a rectangle 13 m by 9 m?	23
Remediation protocol	23
PYQS AND ANSWER PRACTICE	23
Verified 2024-2026 Set-A demand and key ledger	23
Timed mixed transfer	24
Q33. Practice variant 5: A finishes a job in 16 days and B in 22 days. Working together, about how many days do they take (nearest day)?	24
Q34. Practice variant 5: A pipe fills a tank in 14 hours and a leak empties it in 28 hours. How many hours together (nearest hour)?	25
Q35. Practice variant 5: A vehicle travels at 65 km/h for 4 hours. What distance does it cover?	25
Q36. Practice variant 5: A train 160 m long crosses a pole at 54 km/h. How many seconds does it take?	25
Q37. Practice variant 5: A boat moves at 16 km/h in still water and the stream is 2 km/h. What is its downstream speed?	25
Q38. Practice variant 5: What is the smaller angle between clock hands at 6:20?	25
Q39. Practice variant 5: If today is Monday, what weekday is it after 104 days?	25
Q40. Practice variant 5: What is the area in square metres of a rectangle 16 m by 12 m?	25
OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER	25
Rates, Motion, Time and Geometry - ADVANCED	25
1. Architecture	25
2. Advanced tools with conditions	26
3. Harder methods (worked)	26
4. Time-saving techniques (safe conditions)	27
5. Boundary cases	28
6. Advanced traps	28

7. Error analysis	28
8. Advanced drill (with full solutions)	29
9. Timed transfer and retry gate	30
10. PYQ-pattern notes (2024-2026, Set A)	30
11. Study links	30
Historical PYQ Integration (2018-2023)	30
CONSOLIDATED REGISTER NOTES	35
Complete revision spine	35
Ownership and close-option firewall	35
Timed answer route	35

Rates-Motion-Time-and-Geometry - Complete CSAT Learning Session

Identity: csat-04: learner-v2:g1 **Generation date:** 2026-09-06 **Approval:** false **Official syllabus anchor:** Basic numeracy and general mental ability at Class X level.

Source	SHA-256 at generation
upsc-ai-kit\knowledge\CSAT\basic\04_Rates-Motion-Time-and-Geometry.md	5de683879bab3ffad56f2998e92effef79949e7902f4caadadaab6e3c0791b75
upsc-ai-kit\knowledge\CSAT\advanced\04_Rates-Motion-Time-and-Geometry.md	e8eded82a583d79c3550851662fa1c13a53540a0f9b18e7c2420f12179d42a4f
upsc-ai-kit\knowledge\CSAT\00_Master-Framework.md	aaa2016904629d268db79a5798150ef8b9a1386fd1edb2ec737062f2cc9e3fcf
upsc-ai-kit\knowledge\CSAT\OFFICIAL-UPSC-SYLLABUS-MAPPING.md	80dd81eb9ccd7e977570eb86cac69418178dc665d981094c6438a95ca4b8b8a8
upsc-ai-kit\knowledge\CSAT\00_Question-Audit-Ledger.md	fbecba6a750ae88aba541dd1ef1379e6e73841ac7a4285b99bec8fd59bea7e6f

The canonical Basic owner is taught first. Optional Advanced material is isolated after practice. All official-PYQ references preserve the repository's supplied/provisional key labels; no unavailable wording, official explanation or official key is invented.

BASIC LEARNING SESSION

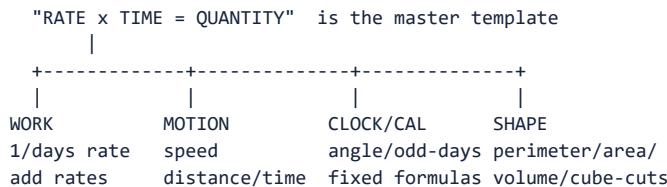
Twelve-panel ASCII master flow

+-----+-----+	
	PANEL 01 - RATE MODEL
	Quantity equals rate multiplied by time.
+-----+-----+	
	v
+-----+-----+	
	PANEL 02 - TIME AND WORK
	Add work rates, not completion times.
+-----+-----+	
	v
+-----+-----+	
	PANEL 03 - PIPES AND LEAKS
	Treat filling positive and emptying negative.
+-----+-----+	
	v
+-----+-----+	
	PANEL 04 - SPEED AND DISTANCE
	Match units before using distance equals speed times time.
+-----+-----+	
	v
+-----+-----+	
	PANEL 05 - RELATIVE MOTION
	Add opposite-direction speeds and subtract same-direction speeds.
+-----+-----+	
	v
+-----+-----+	
	PANEL 06 - TRAINS AND BOATS
	Include object length and current speed correctly.
+-----+-----+	
	v
+-----+-----+	
	PANEL 07 - RACES AND TRACKS
	Use relative distance on linear or circular paths.
+-----+-----+	
	v
+-----+-----+	
	PANEL 08 - CLOCKS
	Use the 5.5-degree-per-minute relative hand speed.
+-----+-----+	
	v
+-----+-----+	
	PANEL 09 - CALENDARS
	Reduce day shifts modulo seven and handle leap-year rules.
+-----+-----+	
	v
+-----+-----+	
	PANEL 10 - GEOMETRY
	Use properties before coordinates or formulas.
+-----+-----+	
	v
+-----+-----+	
	PANEL 11 - MENSURATION
	Distinguish length, area and volume units.
+-----+-----+	
	v
+-----+-----+	
	PANEL 12 - VERIFICATION
	Check dimensions, magnitude, endpoints and physical possibility.
+-----+-----+	

Rates, Motion, Time and Geometry - MUST-DO

Subject: CSAT | **Tier:** Must-Do (foundation) | **Family:** Basic numeracy / general mental ability. **Core skill:** time-and-work, pipes, speed-distance, trains/boats, clocks, calendars and mensuration - all as **rate x time** or a **geometric measure**. **Grounded in:** audited UPSC CSAT PYQ sets 2024-2026 (Set A). Drills are **computationally verified**. **FACT:** = rule/fact | **ANALYSIS:** = guidance | □ = original example. *Companion: Advanced Drill.*

1. Visual foundation



Core proposition: work, pipes and motion are the **same equation** (rate x time = amount); clocks, calendars and mensuration are **fixed-formula** families.

2. Essential formulas

Family	Formula	Edge condition		
FACT: Work	If A finishes in a days, rate = $1/a$; combined rate = sum of rates; time = $1/\text{combined}$.	Rates add only for simultaneous work.		
FACT: Pipes	Fillers add (+1/time), drains subtract (-1/time).	If the net rate is ≤ 0 the tank never fills.		
FACT: Speed	Speed = Distance / Time; $\text{km}/\text{h} \times 5/18 = \text{m}/\text{s}$.	Speed uses path length travelled , never net displacement.		
FACT: Average speed	Total distance / total time. For two equal-distance legs this is the harmonic mean $2s_1s_2/(s_1 + s_2)$.	The harmonic form fails for unequal distances - go back to total/total.		
FACT: Trains	Cross a pole: length/speed. Cross a platform: (length + platform)/speed.	Always include the train's own length.		
FACT: Boats/streams	Downstream speed = $b+s$; upstream speed = $b-s$.	b is still-water speed and must exceed stream speed for upstream progress.		
FACT: Relative speed	Opposite directions add speeds; same direction subtract.	Requires motion along the same line/track .		
FACT: Circular track	First meeting anywhere: $L/(v_1+v_2)$ opposite directions, L/v_1	v_1-v_2	same direction.	Same start point and constant speeds. A start-point reunion requires both runners to complete whole laps; use an LCM only for commensurable lap times.
FACT: Clock angle	Angle $= 30H - 5.5M $	$30H - 5.5M$	degrees.	H is the hour on a 12-hour dial (use $H \bmod 12$, so 14:00 $\rightarrow H = 2$); if the result exceeds 180, the smaller angle is $360 - \text{angle}$.

Family	Formula	Edge condition		
FACT: Calendar	Odd days decide the weekday; a non-leap year shifts the weekday by 1, a leap year by 2.	A Gregorian year is leap iff divisible by 4, except a century must also be divisible by 400; count a +2 shift only when the span contains 29 February.		
FACT: Mensuration	Rectangle: area = $l \times b$, perimeter = $2(l + b)$, diagonal = $\sqrt{l^2 + b^2}$. Triangle: area = $\frac{bh}{2}$; trapezium with parallel sides a, b: area = $\frac{(a+b)h}{2}$; circle: area = πr^2 , circumference = $2\pi r$. Cuboid: volume = lbh , surface = $2(lb+bh+hl)$; cube: volume = a^3 , surface = $6a^2$; right circular cylinder: volume = $\pi r^2 h$; right circular cone: volume = $\frac{\pi r^2 h}{3}$; sphere: volume = $\frac{4\pi r^3}{3}$.	Keep every length in one unit. Area is in square units and volume in cubic units. In a trapezium, h is perpendicular to the parallel sides; cylinder/cone formulas require the radius and perpendicular height. Use $\pi = \frac{22}{7}$ only when the data permit it.		

MEMORY: **Unit reflex:** convert km/h to m/s with $\times \frac{5}{18}$ **before** touching train/length problems - half of all motion errors are unit slips.

2.1 Speed vs velocity - distance vs displacement

Quantity	Definition	Uses
FACT: Distance	Total path length covered, always ≥ 0 and never decreasing.	Fuel, fare, time-taken, average speed.
FACT: Displacement	Net change of position, with direction; can be zero after a round trip.	"How far is he from the start?", forward-and-back step items.
FACT: Speed	distance / time - a magnitude only.	Every "average speed" question in this paper.
FACT: Velocity	displacement / time - carries a direction.	Items asking for net position or bearing.

- ANALYSIS: **The audited papers exploit this.** The 2026 Set-A paper contains a forward/backward step item in which a toy makes a fixed number of jumps of two different lengths and ends a stated distance **forward**: the total path covered and the net advance are completely different numbers, and only the net advance is asked. Read whether the stem wants **how far travelled** or **how far from the start**.

- MEMORY: **Test:** if the object ever reverses, the two answers differ. If it never reverses, they coincide.

3. Method

1. Identify the family (work / motion / clock / calendar / shape).
2. Write **rates** (for work/motion) or plug the **fixed formula** (clock/calendar/shape).
3. For "together" or "relative" problems, **add or subtract rates/speeds** first, then invert.
4. Verify units and magnitude (time positive; distance sensible; angle ≤ 180 typically taken as the smaller angle).

4. Conditions to respect

- ANALYSIS: Add work-rates only when workers **work simultaneously**; for alternate-day work, sum one full cycle first.
- ANALYSIS: Relative speed **adds** for opposite directions and **subtracts** for the same direction - and the objects must be moving along the **same line** for the simple rule.
- ANALYSIS: For a circular track, read "**meet anywhere**" versus "**meet at the starting point**". The latter is a common-whole-laps condition, not the relative-speed meeting time.
- ANALYSIS: The clock formula takes H on a **12-hour dial** and gives an angle; if it exceeds 180, take $360 - \text{angle}$ for the smaller angle.
- ANALYSIS: **Average speed is never the plain mean of the speeds** unless the **times** are equal. For equal **distances** use the harmonic mean; otherwise compute total distance over total time.
- ANALYSIS: Distinguish **distance travelled** from **displacement** before writing any equation (Section 2.1).
- ANALYSIS: For a calendar question, test the **year first**: 2000 is leap, but 1900 and 2100 are not.
- ANALYSIS: In tile-packing, divide the available dimensions by the tile dimensions in a single orientation; rotate only if the stem permits it, and take whole tiles only.

5. Original solved examples

□ Example A (work)

A finishes a job in 12 days, B in 18. Together? Combined rate = $1/12 + 1/18 = 5/36$; time = $36/5 = 7.2$ days. (Verified.)

□ Example B (pipes)

Pipe A fills a tank in 6 h, B in 8 h. Together? $1/6 + 1/8 = 7/24 \rightarrow$ time = $24/7 \approx 3.43$ h. (Verified.) If instead A fills in 4 h and a drain empties in 6 h: $1/4 - 1/6 = 1/12 \rightarrow 12$ h.

□ Example C (train)

A 120 m train at 72 km/h (= 20 m/s). Crosses a pole: $120/20 = 6$ s. Crosses a 180 m platform: $(120 + 180)/20 = 15$ s. (Verified.)

□ Example C.1 (boat and circular track)

A boat has still-water speed 10 km/h and stream speed 2 km/h. Its downstream and upstream speeds are **12 km/h** and **8 km/h** respectively; reversing the signs is the standard trap.

On a 400 m track, runners at 5 m/s and 3 m/s starting together meet in the same direction after $400/|5-3| = 200$ s. Opposite-direction meeting would be $400/(5+3) = 50$ s. (Verified.)

□ Example D (clock)

Angle between the hands at 4:20. $|30 \times 4 - 5.5 \times 20| = |120 - 110| = 10$ degrees. (Verified.) At 2:50 the same formula gives $|60 - 275| = 215$, which is more than 180, so the smaller angle is $360 - 215 = 145$ degrees. (Verified.)

□ Example E (mensuration)

Rectangle 8 x 6. Area = 48, perimeter = 28, diagonal = $\sqrt{64 + 36} = 10$. (Verified.)

Geometry transfer. A right triangle has legs 6 cm and 8 cm. Its hypotenuse is $\sqrt{6^2 + 8^2} = 10$ cm; its area is $6 \times 8 / 2 = 24$ cm². A circle of radius 7 cm has area $(22/7) \times 7^2 = 154$ cm². (Verified.) Use Pythagoras only after identifying a right angle.

Mensuration transfer. A trapezium with parallel sides 8 cm and 12 cm and perpendicular height 5 cm has area $(8 + 12) \times 5 / 2 = 50$ cm². A right circular cylinder of radius 3 cm and height 10 cm has volume $\pi \times 3^2 \times 10 = 90\pi$ cm³; the cone with the same base and height has one-third of that volume, 30π cm³. A sphere of radius 3 cm has volume $4\pi \times 3^3 / 3 = 36\pi$ cm³. (Verified.)

□ Example E.1 (tile packing)

How many 20 cm by 10 cm tiles fit in a 1 m by 80 cm rectangle without cutting? Convert first: 1 m = 100 cm. Along the sides, $100/20 = 5$ and $80/10 = 8$, hence **40 tiles**. (*Verified.*) The calculation would be invalid if the dimensions were mixed or cutting were allowed.

□ Foundation cube count (needed before the drill)

For a **fully painted solid** $n \times n \times n$ cube cut along all grid planes:

Location	Count	Why
Corners: 3 painted faces	8	A cube has eight corners.
Edge interiors: exactly 2 faces	$12(n-2)$	Twelve edges, with $n-2$ non-corner cubes each.
Face interiors: exactly 1 face	$6(n-2)^2$	Six faces, excluding edges.
Inner cubes: 0 faces	$(n-2)^3$	Remove one painted layer from each side.

These counts require integer $n \geq 2$, a solid cube, and paint on all six outer faces. They sum to n^3 ; that is the check before using them in the mini-drill. The Advanced companion extends, rather than supplies, this foundation.

□ Example F (distance vs displacement, and average speed)

A cyclist rides 12 km east in 30 min, then 12 km back west in 20 min.

- **Distance** covered = 24 km; **displacement** = 0 (he is back where he started).
- **Average speed** = $24 \text{ km} / (50/60 \text{ h}) = \mathbf{28.8 \text{ km/h}}$.
- **Average velocity** = $0 / \text{time} = \mathbf{0}$.
- The plain mean of the leg speeds (24 and 36 km/h) is 30 - **wrong**. The harmonic mean

$2 \times 24 \times 36 / (24 + 36) = 1728/60 = \mathbf{28.8}$, which matches, because the two legs are equal in **distance**. (*Verified.*)

6. Must-Know facts

- FACT: $\text{km/h} \times 5/18 = \text{m/s}$; $\text{m/s} \times 18/5 = \text{km/h}$.
- FACT: The clock hands **coincide 11 times** every 12 hours (22 times a day), not 12 - once every $720/11 = 65 \frac{5}{11}$ minutes.
- FACT: The hands are at **right angles 22 times** in 12 hours (44 times a day).
- FACT: 100 years contain **5 odd days**; 400 years contain **0** (the weekday pattern repeats every 400 years).
- FACT: A Gregorian century year is leap only when divisible by 400: 2000 is leap; 1900 and 2100 are not.
- FACT: Cube side doubled $\rightarrow$ surface $\times 4$, volume $\times 8$.
- FACT: In a fully painted $n \times n \times n$ cube, exactly-two-face cubes are $12(n-2)$; for $3 \times 3 \times 3$, that is 12.
- FACT: For same-direction motion, the **faster catches the slower** at the difference of speeds.
- FACT: **Displacement is never greater than distance**, and they are equal only when the motion never reverses.

7. Common traps

- WRONG: Adding times instead of rates for "working together". -> Add **rates**, then invert.
- WRONG: Forgetting the train's **own length** when crossing a platform/bridge. -> Add both lengths.
- WRONG: Using 30H alone for the clock angle. -> Include the minute term $-5 \cdot 5M$.
- WRONG: Feeding a 24-hour hour into the clock formula. -> Use $H \bmod 12$ (17:40 -> $H = 5$).
- WRONG: Assuming hands coincide 12 times in 12 hours. -> They coincide **11** times.
- WRONG: Averaging two speeds by a plain mean for a round trip. -> Use the harmonic mean $2 \times s_1 \times s_2 / (s_1 + s_2)$, and only when the two legs are **equal in distance**.
- WRONG: Answering with the **distance covered** when the stem asks how far the object ends up ****from the start****. -> That is displacement.
- WRONG: Assuming a leap-year span always shifts the weekday by 2. -> Only if 29 February falls inside it.
- WRONG: Calling every year divisible by 4 a leap year. -> Apply the century/400 exception before counting odd days.
- WRONG: Treating a tile-area quotient as a packing count when side lengths do not fit. -> Check both dimensions and whether rotation/cutting is allowed.

8. Quick checks

- FACT: Can you convert 90 km/h to m/s in one step? (25 m/s.)
- FACT: Can you write the combined rate for two workers instantly?
- FACT: Can you state the clock angle formula without hesitation - including the $H \bmod 12$ rule?
- FACT: Can you say, for a there-and-back trip, what the displacement is without computing anything? (Zero.)

9. Mini-drill (with answers and explanations)

1. A is twice as fast as B; together they finish in 8 days. How long does A alone take?
2. A 150 m train crosses a 350 m bridge in 20 s. Find its speed.
3. A $3 \times 3 \times 3$ painted cube is cut into 27 unit cubes. How many have **exactly two** painted faces?
4. If 15 August 2024 (a leap year) is a Thursday, what day is 15 August 2025?
5. Between 4 and 5 o'clock, at what minute do the hands coincide?
6. A grasshopper makes 20 hops in a straight line: each forward hop is 6 cm, each backward hop is 4 cm, and it ends up 40 cm ahead of its start. How many forward hops did it make, and what total distance did it cover?
7. A car covers 60 km at 30 km/h and the next 60 km at 60 km/h. Find its average speed.

Answers.

1. **12 days.** A's rate = $2B$; combined = $3B = 1/8 \rightarrow B = 1/24, A = 1/12$. *(Verified: $1/12 + 1/24 = 1/8$.)*
2. **25 m/s (90 km/h).** $(150 + 350)/20 = 25 \text{ m/s}$. (Verified.)
3. **12.** Exactly-two-face cubes lie on edges: $12 \times (3 - 2) = 12$. (Verified.)
4. **Friday.** 15 Aug 2024 -> 15 Aug 2025 spans 365 days (Feb 2025 is non-leap); $365 \bmod 7 = 1 \rightarrow \text{Thursday} + 1 = \text{Friday}$. (Verified.)
5. **21 and 9/11 minutes (about 4:21:49).** Coincidence minute = $60H/11 = 240/11 \approx 21.82$. (Verified.)
6. **12 forward hops; 104 cm of distance covered.** Let f forward and $b = 20 - f$ backward.
Displacement: $6f - 4(20 - f) = 40 \rightarrow 10f - 80 = 40 \rightarrow f = 12$, so $b = 8$. **Distance (path length):** $6 \times 12 + 4 \times 8 = 72 + 32 = 104 \text{ cm}$, while the displacement is only **40 cm**. (Verified: $72 - 32 = 40$.) ANALYSIS: Two different correct numbers from one motion - read which the stem wants.

7. **40 km/h.** Total distance 120 km; total time $60/30 + 60/60 = 2 + 1 = 3$ h; $120/3 = 40$.

The plain mean of 30 and 60 is 45 - **wrong**. The harmonic mean applies here because the legs are equal in distance: $2 \times 30 \times 60 / (30 + 60) = 3600/90 = 40$. (Verified.)

10. Timed transfer, diagnosis and retry gate

Set a 90-second ceiling per item. Park an item at the ceiling; in a qualifying paper, a clean next question is worth more than a heroic calculation.

1. If 1 March 2100 is Monday, what day is 1 March 2101?
2. A $4 \times 5 \times 6$ cm cuboid is painted on every outer face and cut into 1 cm cubes. How many cubes have exactly three painted faces?
3. A $10 \text{ m} \times 8 \text{ m}$ floor is tiled by $2 \text{ m} \times 1 \text{ m}$ tiles without cutting. How many tiles are required?
4. A 180 m train at 54 km/h crosses a pole. Find the time.
5. A boat travels 24 km downstream and 24 km upstream at still-water speed 8 km/h in a 2 km/h stream. Find the total time.
6. Two runners at 7 m/s and 3 m/s start together on a 200 m track in the same direction. When do they first meet anywhere?

Answers. 1. **Tuesday**: 2100 is not leap, so the span has 365 days. 2. **8**: every cuboid has eight corners. 3. **40**. 4. **12 s**: $54 \text{ km/h} = 15 \text{ m/s}$, then $180/15$. 5. **6.4 h**: $24/10 + 24/6 = 2.4 + 4$. 6. **50 s**: $200 / |7-3|$.

If you missed because...	Code	Repair before retry
You counted 2100 as leap	C/R	Write $\div 4$; century - $\blacksquare >$ $\div 400$ above the calendar.
You used area alone for tiles	A	Draw the two side quotients before multiplying.
You confused cube faces/corners	C	Rebuild the four-row cube table and check its total.
You left speed in km/h	X	Put every train speed in m/s before division.

Retry gate: redo the six items after a 20-minute gap. Continue only at **6/6 within nine minutes** and with no unresolved C or R error; otherwise redo the matching worked method, then retry a fresh six-item set.

11. Study links

- FACT: Optional Advanced companion - alternate-day work, multi-segment comparisons, and harder cube/circular cases. Core above is sufficient to practise the basic paper forms.
- FACT: Arithmetic and Commercial Math - ratios/averages feed speed and work.
- FACT: Number Systems and Number Sense - odd-days counting is modular arithmetic.

2026 PYQ Integration

Status: 2026 question-level PYQ demand is integrated into this owner. **Provenance**: Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2026.md. **Answer-key rule**: The 2026 Prelims and CSAT Set-A keys held locally are **provisional**; no option or answer is recorded or inferred in this integration.

- **Year represented**: 2026
- **Paper(s)**: CSAT
- **Routed question demands**: 8

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	CSAT	13	Net displacement with jumps	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	20	Relative-speed train crossing	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	24	Cube-cutting sequence count	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	37	Circular-track relative motion	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	41	Sound-distance relative rate	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	50	Multi-segment speed comparison	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	CSAT	68	Rectangular-tile area count	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	69	Train-accident relative rate	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- Net displacement with jumps
- Relative-speed train crossing
- Cube-cutting sequence count
- Circular-track relative motion
- Sound-distance relative rate
- Multi-segment speed comparison
- Rectangular-tile area count
- Train-accident relative rate

This block integrates the 2026 examinable demand and paper metadata. It is kept separate from the 2018-2023 and 2024-2025 blocks and does not convert a provisionally-keyed, answer-free objective question into a solved answer.

Recent PYQ Integration (2024-2025)

Status: 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2024-2025.md. **Answer-key rule:** The official 2024-2025 Prelims Set-A keys are present in the repository and CSAT Set-A keys are supplied; even so, no option or answer is recorded or inferred in this integration.

- **Years represented:** 2024, 2025
- **Paper(s):** CSAT
- **Routed question demands:** 13

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	CSAT	5	Cube partition cuts	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	CSAT	7	Work-rate scaling	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	8	Alternating-worker schedule	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	30	Metric-unit conversion	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	37	Clock-hand coincidence	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	38	Calendar repetition	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	40	Clock-hand angle	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	9	Run-rate time relation	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	15	Painted-cube partition	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	37	Circular-track encounters	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	39	Overtaking relative speed	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2025	CSAT	45	Pipes-and-cistern rate	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	59	Fractional work rates	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- Cube partition cuts
- Work-rate scaling
- Alternating-worker schedule
- Metric-unit conversion
- Clock-hand coincidence
- Calendar repetition
- Clock-hand angle
- Run-rate time relation
- Painted-cube partition
- Circular-track encounters
- Overtaking relative speed
- Pipes-and-cistern rate
- Fractional work rates

This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2018-2023.md. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2019, 2020, 2021, 2022, 2023
- **Paper(s):** CSAT
- **Routed question demands:** 45

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	5	Velocity-time distance ratio	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	6	Train crossing time	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	35	Painted-cube face geometry	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	40	Circular track speed ratio	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	58	Inscribed triangle count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	59	Calendar date sequence	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	60	Rectangle diagonal geometry	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	12	Calendar day determination	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	16	Race distance inference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	30	Right-triangle distance	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	35	LCM cycle coincidence	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	36	Race start scaling	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	49	Fast clock time correction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	57	Path displacement distance	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	59	Calendar year repeat	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	20	Calendar day calculation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2020	CSAT	31	Compass direction tracking	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	38	Cube volume density ratio	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	71	Combined work-rate fraction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	72	Multi-leg average speed	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	73	Circle-square intersection count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	74	Upstream-downstream speed ratio	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	80	Jump-sequence well climb	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	16	Bounce count sequence	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	18	Direction-distance path	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	20	Net displacement direction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	29	Two walkers meeting point	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	48	Calendar day calculation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	57	Clock-hand angle timing	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	63	Collinear-points ratio count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2021	CSAT	68	Volume unit conversion	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	75	LCM calendar meeting day	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	80	Work-rate remaining days	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	6	Direction distance tracking	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	7	Calendar day finding	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	15	Circular track overtaking	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	35	Time unit total conversion	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	46	Clock route time comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	57	Work rate comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	59	Rectangle cut feasibility	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	68	Clock hand coincidence	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	20	Cyclic work-rate schedule	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	49	Day-of-week calculation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	50	Periodic signal synchronisation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2023	CSAT	67	Tile packing maximum	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- Velocity-time distance ratio
- Train crossing time
- Painted-cube face geometry
- Circular track speed ratio
- Inscribed triangle count
- Calendar date sequence
- Rectangle diagonal geometry
- Calendar day determination
- Race distance inference
- Right-triangle distance
- LCM cycle coincidence
- Race start scaling
- Fast clock time correction
- Path displacement distance
- Calendar year repeat
- Calendar day calculation
- Compass direction tracking
- Cube volume density ratio
- Combined work-rate fraction
- Multi-leg average speed
- Circle-square intersection count
- Upstream-downstream speed ratio
- Jump-sequence well climb
- Bounce count sequence
- Direction-distance path
- Net displacement direction
- Two walkers meeting point
- Clock-hand angle timing
- Collinear-points ratio count
- Volume unit conversion
- LCM calendar meeting day
- Work-rate remaining days
- Direction distance tracking
- Calendar day finding
- Circular track overtaking
- Time unit total conversion
- Clock route time comparison
- Work rate comparison

- Rectangle cut feasibility
- Clock hand coincidence
- Cyclic work-rate schedule
- Day-of-week calculation
- Periodic signal synchronisation
- Tile packing maximum

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

Semantic-completeness closure - 2026-09-06

Literal syllabus and canonical ownership

- **Literal clause:** Basic numeracy and general mental ability at Class X level.
- **Canonical scope:** Owns time-work, pipes, speed-distance, relative speed, trains, boats, races, clocks, calendars, elementary geometry, mensuration and unit conversion.
- **Cross-topic boundary:** Commercial arithmetic belongs to Topic 03; algebraic comparison and sufficiency to Topic 05; spatial direction puzzles to Topic 06.

Complete learner route

1. **Rate model:** Quantity equals rate multiplied by time.
2. **Time and work:** Add work rates, not completion times.
3. **Pipes and leaks:** Treat filling positive and emptying negative.
4. **Speed and distance:** Match units before using distance equals speed times time.
5. **Relative motion:** Add opposite-direction speeds and subtract same-direction speeds.
6. **Trains and boats:** Include object length and current speed correctly.
7. **Races and tracks:** Use relative distance on linear or circular paths.
8. **Clocks:** Use the 5.5-degree-per-minute relative hand speed.
9. **Calendars:** Reduce day shifts modulo seven and handle leap-year rules.
10. **Geometry:** Use properties before coordinates or formulas.
11. **Mensuration:** Distinguish length, area and volume units.
12. **Verification:** Check dimensions, magnitude, endpoints and physical possibility.

Verification and hostile-query gate

Rate equations retain units. Geometry formulas are derived or decomposed, and generated solutions are checked dimensionally and by direct substitution.

The hostile absence search explicitly tested these families and close-option terms: **time and work; pipes; relative speed; train; boat; clock; calendar; mensuration**. A shortcut is usable only when its stated condition survives; otherwise return to the first-principles representation. Every worked answer must finish with an independent check: passage support, substitution, enumeration, units, bounds, reverse operation or option elimination.

Difficulty and timed progression

1. Foundation: recognise the family and state the governing definition or relation.
2. Core: solve a direct item with a visible representation and one verification.
3. Advanced: combine two mechanisms, test edge cases and reject close distractors.
4. Timed: use classify -> extract -> represent -> execute -> verify -> decide.
5. Remediation: log the error as concept, application, calculation, reading, passage, time or guess; return to the owning subtopic before retesting.

BASIC MCQS / REMEDIATION

Diagnostic and core set

Q1. Practice variant 1: A finishes a job in 12 days and B in 18 days. Working together, about how many days do they take (nearest day)?

A. 7 B. 8 C. 6 D. 9

Correct answer: A. Add work rates $1/d1 + 1/d2$, then invert.

Q2. Practice variant 1: A pipe fills a tank in 10 hours and a leak empties it in 20 hours. How many hours together (nearest hour)?

A. 21 B. 20 C. 19 D. 22

Correct answer: B. Net rate equals fill rate minus leak rate.

Q3. Practice variant 1: A vehicle travels at 45 km/h for 4 hours. What distance does it cover?

A. 181 B. 179 C. 180 D. 182

Correct answer: C. Distance = speed x time after matching units.

Q4. Practice variant 1: A train 120 m long crosses a pole at 54 km/h. How many seconds does it take?

A. 9 B. 7 C. 10 D. 8

Correct answer: D. Convert km/h to m/s, then time = train length/speed.

Q5. Practice variant 1: A boat moves at 12 km/h in still water and the stream is 2 km/h. What is its downstream speed?

A. 14 B. 15 C. 13 D. 16

Correct answer: A. Downstream speed equals still-water speed plus stream speed.

Q6. Practice variant 1: What is the smaller angle between clock hands at 2:20?

A. 51 B. 50 C. 49 D. 52

Correct answer: B. Hour hand angle is $30h + 0.5m$; minute hand angle is $6m$.

Q7. Practice variant 1: If today is Monday, what weekday is it after 100 days?

A. Tuesday B. Thursday C. Wednesday D. Friday

Correct answer: C. Weekdays repeat modulo seven.

Q8. Practice variant 1: What is the area in square metres of a rectangle 12 m by 8 m?

A. 97 B. 95 C. 98 D. 96

Correct answer: D. Rectangle area = length x breadth; the unit is squared.

Q9. Practice variant 2: A finishes a job in 13 days and B in 19 days. Working together, about how many days do they take (nearest day)?

A. 8 B. 9 C. 7 D. 10

Correct answer: A. Add work rates $1/d1 + 1/d2$, then invert.

Q10. Practice variant 2: A pipe fills a tank in 11 hours and a leak empties it in 22 hours. How many hours together (nearest hour)?

A. 23 B. 22 C. 21 D. 24

Correct answer: B. Net rate equals fill rate minus leak rate.

Q11. Practice variant 2: A vehicle travels at 50 km/h for 4 hours. What distance does it cover?

A. 201 B. 199 C. 200 D. 202

Correct answer: C. Distance = speed x time after matching units.

Q12. Practice variant 2: A train 130 m long crosses a pole at 54 km/h. How many seconds does it take?

A. 10 B. 8 C. 11 D. 9

Correct answer: D. Convert km/h to m/s, then time = train length/speed.

Q13. Practice variant 2: A boat moves at 13 km/h in still water and the stream is 2 km/h. What is its downstream speed?

A. 15 B. 16 C. 14 D. 17

Correct answer: A. Downstream speed equals still-water speed plus stream speed.

Q14. Practice variant 2: What is the smaller angle between clock hands at 3:20?

A. 21 B. 20 C. 19 D. 22

Correct answer: B. Hour hand angle is $30h + 0.5m$; minute hand angle is $6m$.

Q15. Practice variant 2: If today is Monday, what weekday is it after 101 days?

A. Tuesday B. Wednesday C. Thursday D. Friday

Correct answer: C. Weekdays repeat modulo seven.

Q16. Practice variant 2: What is the area in square metres of a rectangle 13 m by 9 m?

A. 118 B. 116 C. 119 D. 117

Correct answer: D. Rectangle area = length x breadth; the unit is squared.

Remediation protocol

1. Recompute without options.
2. Name the failed rule or passage phrase.
3. Reject every distractor for a specific reason.
4. Retry with altered numbers, wording or constraints.
5. Advance only after two consecutive correct answers under the time ceiling.

PYQS AND ANSWER PRACTICE

Verified 2024-2026 Set-A demand and key ledger

Year	Q	Neutral verified demand	Key status	Solution architecture
2024	5	Cube partition cuts	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	7	Work-rate scaling	C (supplied)	Add rates and invert the net rate.
2024	8	Alternating-worker schedule	C (supplied)	Add rates and invert the net rate.

Year	Q	Neutral verified demand	Key status	Solution architecture
2024	30	Metric-unit conversion	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	37	Clock-hand coincidence	A (supplied)	Use relative angular speed.
2024	38	Calendar repetition	C (supplied)	Reduce the day shift modulo seven.
2024	40	Clock-hand angle	C (supplied)	Use relative angular speed.
2025	9	Run-rate time relation	D (supplied)	Draw a labelled generation graph.
2025	15	Painted-cube partition	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	37	Circular-track encounters	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	39	Overtaking relative speed	C (supplied)	Use relative speed after unit conversion.
2025	45	Pipes-and-cistern rate	D (supplied)	Assign positive fill and negative empty rates.
2025	59	Fractional work rates	B (supplied)	Add rates and invert the net rate.
2026	13	Net displacement with jumps	D (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	20	Relative-speed train crossing	C (provisional)	Use relative speed after unit conversion.
2026	24	Cube-cutting sequence count	A (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	37	Circular-track relative motion	A (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	41	Sound-distance relative rate	C (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	50	Multi-segment speed comparison	D (provisional)	Use relative speed after unit conversion.
2026	68	Rectangular-tile area count	A (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	69	Train-accident relative rate	A (provisional)	Use the total crossing length.

The table is a non-verbatim routing audit. It records the locally checked Set-A key letter and a valid solving route, but does not pretend that UPSC publishes model solutions. The separate workbook supplies original solved equivalents for every mechanism.

Timed mixed transfer

Q33. Practice variant 5: A finishes a job in 16 days and B in 22 days. Working together, about how many days do they take (nearest day)?

A. 9 B. 10 C. 8 D. 11

Correct answer: A. Add work rates $1/d_1 + 1/d_2$, then invert.

Q34. Practice variant 5: A pipe fills a tank in 14 hours and a leak empties it in 28 hours. How many hours together (nearest hour)?

A. 29 B. 28 C. 27 D. 30

Correct answer: B. Net rate equals fill rate minus leak rate.

Q35. Practice variant 5: A vehicle travels at 65 km/h for 4 hours. What distance does it cover?

A. 261 B. 259 C. 260 D. 262

Correct answer: C. Distance = speed x time after matching units.

Q36. Practice variant 5: A train 160 m long crosses a pole at 54 km/h. How many seconds does it take?

A. 12 B. 10 C. 13 D. 11

Correct answer: D. Convert km/h to m/s, then time = train length/speed.

Q37. Practice variant 5: A boat moves at 16 km/h in still water and the stream is 2 km/h. What is its downstream speed?

A. 18 B. 19 C. 17 D. 20

Correct answer: A. Downstream speed equals still-water speed plus stream speed.

Q38. Practice variant 5: What is the smaller angle between clock hands at 6:20?

A. 71 B. 70 C. 69 D. 72

Correct answer: B. Hour hand angle is $30h + 0.5m$; minute hand angle is $6m$.

Q39. Practice variant 5: If today is Monday, what weekday is it after 104 days?

A. Tuesday B. Wednesday C. Sunday D. Thursday

Correct answer: C. Weekdays repeat modulo seven.

Q40. Practice variant 5: What is the area in square metres of a rectangle 16 m by 12 m?

A. 193 B. 191 C. 194 D. 192

Correct answer: D. Rectangle area = length x breadth; the unit is squared.

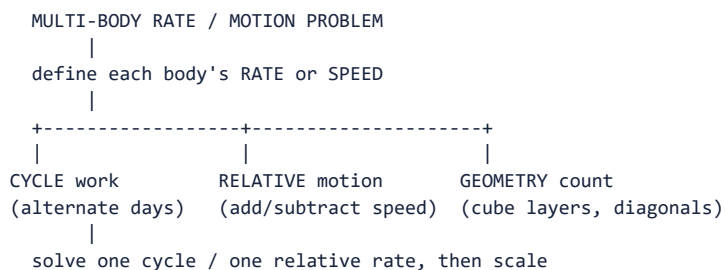
OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

Rates, Motion, Time and Geometry - ADVANCED

Subject: CSAT | **Tier:** Advanced Drill | **Family:** Basic numeracy / general mental ability. **Core skill:** alternate-day work, boats/streams, circular-track meetings, moving-object crossings, clock coincidences and cube-cutting counts.

Grounded in: audited UPSC CSAT PYQ sets 2024-2026 (Set A). Drills are **computationally verified**. **FACT:** = rule/fact | **ANALYSIS:** = guidance | $\square$ = original example. *Companion: Foundation.*

1. Architecture



Analytical claim: rate and relative-motion items reduce to a **net rate** (per cycle or relative), after which they become a division. Cube-cutting items are different: they use geometric layer/face-count formulas.

2. Advanced tools with conditions

Tool	Statement	Condition		
FACT: Alternate-day work	Sum a full A+B cycle, count whole cycles, finish the remainder day by day.	Track who works on the final partial day - who starts changes the answer .		
FACT: Boats/streams	Downstream = $b + s$, upstream = $b - s$ (b = still-water, s = stream).	$b > s$ for forward motion.		
FACT: Circular track	Same direction meet time = L/v	$v_1 - v_2$	; opposite = $L/(v_1 + v_2)$	Same start point and constant speeds. A start-point reunion needs $t/\text{LapTime1}$ and $t/\text{LapTime2}$ both integral; use an LCM only when the lap times are commensurable.
FACT: Moving-object crossing	Time = combined length / relative speed.	Add speeds if opposite, subtract if same direction.		
FACT: Multi-segment average speed	total distance / total time - never the mean of the segment speeds.	Two journeys with the same speeds over the same distances can still take different times if the speeds are attached to different segments.		
FACT: Clock right angle	$\backslash$	$30H - 5.5M$	$= 90$ (or 270)	22 right angles per 12 hours; solutions outside $0 \leq M < 60$ belong to the next hour.
FACT: Cube cut into n^3	3-face = 8 ; 2-face = $12(n-2)$; 1-face = $6(n-2)^2$; 0-face = $(n-2)^3$.	Integer $n \geq 2$, a solid cube, equal cuts, and paint on all six faces; the four counts sum to n^3 .		

3. Harder methods (worked)

□ Method 1 - alternate-day work

A alone in 10 days, B alone in 15; they work on alternate days, A first. Total time? One A+B cycle = $1/10 + 1/15 = 1/6$ per 2 days. After 10 days (5 cycles) = $5/6$ done. Day 11 (A) adds $1/10 \rightarrow 14/15$. Day 12 (B) adds $1/15 \rightarrow$ finished. **Total = 12 days.** (Verified: $5/6 \rightarrow 0.9333 \rightarrow 1.0$.)

□ Method 2 - circular track

A track is 400 m; two runners at 5 m/s and 3 m/s from the same point. Same direction meet time = $400/(5 - 3) = 200$ s. Opposite directions = $400/(5 + 3) = 50$ s. (Verified.)

□ Method 3 - moving-object crossing

A 200 m train at 90 km/h (= 25 m/s) overtakes a man running 5 m/s in the same direction. Relative speed = $25 - 5 = 20$ m/s; time = $200/20 = 10$ s. (Verified.)

□ Method 4 - who starts decides the finish (alternate-day escalation)

A alone in 12 days, B alone in 18; they work on alternate days. One cycle = $1/12 + 1/18 = 5/36$ per 2 days, so 7 cycles (14 days) complete $35/36$ either way. The **remaining $1/36$ is done by whoever's turn day 15 is:**

Starter	Day-15 worker	Rate on day 15	Extra time	Total
A first	A	$1/12 = 3/36$ per day	$(1/36)/(3/36) = 1/3$ day	14 $1/3$ days
B first	B	$1/18 = 2/36$ per day	$(1/36)/(2/36) = 1/2$ day	14 $1/2$ days

(Verified by day-by-day simulation: $43/3$ and $29/2$.)

- MEMORY: **Escalation rule:** whenever a cycle repeats, the answer is decided by the **remainder**, not the cycle. Never divide total work by the cycle rate and stop.

□ Method 5 - same speeds, different times (the segment-ordering trap)

A walks 2 km at 4 km/h, then 3 km at 6 km/h, then 4 km at 8 km/h. B walks the same three distances at 8, 6 and 4 km/h respectively. Same total distance (9 km) and the same **set** of speeds:

- A: $2/4 + 3/6 + 4/8 = 0.5 + 0.5 + 0.5 = 1.5$ h = 90 min.
- B: $2/8 + 3/6 + 4/4 = 0.25 + 0.5 + 1.0 = 1.75$ h = 105 min.
- **A finishes 15 minutes earlier.** (Verified.)

- MEMORY: **Why:** time is distance/speed, so a **slow speed on a long segment** dominates. Pairing the fastest speed with the longest segment always wins. The audited 2026 Set-A paper uses exactly this shape - two riders, the same three speeds, the segment order swapped.

□ Method 6 - calendar and dimensional geometry under pressure

Calendar. What weekday is 1 March 2101 if 1 March 2100 is Monday? Year 2100 is a century not divisible by 400, so it is **not leap**. The interval has 365 days, or one odd day: **Tuesday**. Do not treat "divisible by 4" as the whole Gregorian test.

Tile feasibility. A 9 m x 7 m rectangle is to be covered by 3 m x 1 m tiles, with rotation allowed and no cutting. With the 3 m side along 9 m, the count is $3 \times 7 = 21$; after rotation, $7/3$ is not integral, so that orientation fails. Therefore **21 tiles** is feasible only in the first orientation. State the orientation, not merely area/tile area.

4. Time-saving techniques (safe conditions)

- ANALYSIS: **Net-rate-per-cycle** for alternate work. *Safe once you handle the final partial day explicitly.*
- ANALYSIS: **Relative speed** turns two moving bodies into one. *Safe when motion is along the same line or a common track.*
- ANALYSIS: **Cube-layer formulas** give all four face-counts instantly. *Safe for a solid cube cut into equal unit cubes with $n \geq 2$.*
- ANALYSIS: **Harmonic mean** $2 s_1 s_2 / (s_1 + s_2)$ for equal-distance round trips. *Safe only when the two legs cover the same distance;* for three or more segments, or unequal distances, fall back to $\text{total distance} / \text{total time}$.
- ANALYSIS: **Rearrangement check** for segment-ordering comparisons: with a fixed set of distances and a fixed set of speeds, total time is smallest when the **longest** distance is paired with the **fastest** speed. *Safe for any number of segments - it lets you rank two travellers without computing either time in full.*

5. Boundary cases

- ANALYSIS: Alternate-day problems can finish **mid-cycle**; never just divide total work by the cycle rate.

Who starts changes the total (Method 4) even though the cycle rate is identical.

- ANALYSIS: On a circular track, "meet at the starting point" differs from "meet anywhere" - the latter uses relative speed. For a start-point reunion, solve for a common multiple of the lap times; an LCM is valid only when those times are commensurable. Otherwise a finite reunion need not exist.

- ANALYSIS: Cube formulas assume a **fully painted** solid cube; a hollow or partially painted cube breaks them. For $n = 2$ the 1-face and 0-face counts are correctly **zero**.

- ANALYSIS: A clock right-angle equation has **two** roots per hour; discard any root with $M \geq 60$ - it belongs to the next hour.

- ANALYSIS: Average speed and average velocity differ whenever the path reverses; a closed round trip has a positive average speed and **zero** average velocity.

6. Advanced traps

- WRONG: Dividing total work by the cycle rate and ignoring the partial final day. -> Finish day by day.
- WRONG: Assuming the alternate-day total is the same whoever starts. -> The **remainder** is done at the starter's-turn rate; recompute.
- WRONG: Using still-water speed for downstream/upstream. -> Add/subtract the stream speed.
- WRONG: Adding speeds for same-direction overtaking. -> **Subtract** for same direction; add only for opposite.
- WRONG: Averaging the segment speeds of a multi-leg journey. -> total distance / total time.
- WRONG: Reporting the distance travelled when the stem asked how far the object is **from the start**.
-> That is displacement.
- WRONG: Forgetting that an n^3 cube has $(n-2)^3$ fully unpainted inner cubes. -> Use the layer formula and check the four counts sum to n^3 .

7. Error analysis

Recurring miss	Master-Framework code	Fix
Partial-day work	A	Simulate the last cycle explicitly.
Relative-speed sign	C	Write "opposite = add, same = subtract" each time.
Unit conversion	X	Convert to m/s before any length problem.
Cube face-count	C	Memorise the four-count formula; check the sum equals n^3 .
Calendar century rule	R	Test divisibility by 4, then the 100/400 exception.
Tile/geometry feasibility	A	Check side quotients and stated rotation/cutting conditions, not area alone.

8. Advanced drill (with full solutions)

ANALYSIS: These items are **new** - they do not repeat Methods 1-6 above or the Foundation file.

1. A boat's still-water speed is 10 km/h and the stream is 2 km/h. Time for a 48 km each-way round trip.
2. A 120 m train at 54 km/h and a 180 m train at 36 km/h move towards each other on parallel tracks. How long do they take to cross each other completely?
3. A painted cube is cut into $5 \times 5 \times 5 = 125$ unit cubes. How many have 0, 1, 2 and 3 painted faces?
4. At what time just after 3 o'clock are the hands of a clock at right angles?
5. Two runners on a 400 m circular track run at 8 m/s and 5 m/s in the **same** direction from the same point. When do they first meet **anywhere**, and when do they first meet **at the starting point**?
6. A cyclist rides 15 km uphill at 10 km/h and returns down the same road at 30 km/h. Find the average speed for the whole trip, and the average **velocity**.
7. If 1 January 2099 is Friday, what weekday is 1 January 2100? What extra check is required before moving to 2101?
8. A 12 m x 9 m floor is covered by 3 m x 2 m tiles without cutting. Is the stated orientation feasible, and how many tiles are needed if rotation is allowed?

Solutions.

1. **10 hours.** Downstream 12 km/h $\rightarrow 48/12 = 4$ h; upstream 8 km/h $\rightarrow 48/8 = 6$ h; total 10 h. (Verified.)
2. **12 s.** Opposite directions, so **add**: relative speed = 90 km/h = $90 \times 5/18 = 25$ m/s. The crossing covers **both** lengths: $(120 + 180)/25 = 300/25 = 12$ s. (Verified.) ANALYSIS: Subtracting the speeds (the same-direction rule) would give 5 m/s and 60 s - a fivefold error.
3. **3-face 8, 2-face 36, 1-face 54, 0-face 27.** $12(5-2) = 36$, $6(5-2)^2 = 54$, $(5-2)^3 = 27$, corners = 8; sum = $8 + 36 + 54 + 27 = 125$. (Verified - always check the sum equals n^3 .)
4. **32 and 8/11 minutes past 3 (about 3:32:44).** With M minutes past 3, the gap is $|30 \times 3 - 5.5M| = |90 - 5.5M|$. Setting it to 90 gives $M = 0$ (the hands are already at 90 degrees at exactly 3:00) or $5.5M = 180$, i.e. $M = 360/11 = 32 \frac{8}{11}$. The other root, $|90 - 5.5M| = 270$, gives $M = 720/11 \approx 65.45$, which is past 4 o'clock. (Verified.)
5. **Anywhere:** $400/3 \approx 133.3$ s. **At the starting point:** 400 s. Meeting *anywhere* uses the relative speed: $400/(8 - 5) = 400/3$ s. Meeting *at the start* needs both to have completed whole laps: lap times are $400/8 = 50$ s and $400/5 = 80$ s, so the first common moment is $\text{LCM}(50, 80) = 400$ s. (Verified.) ANALYSIS: These are **different questions with different answers** - exactly the boundary case flagged in Section 5.
6. **Average speed 15 km/h; average velocity 0.** Total distance = 30 km; total time = $15/10 + 15/30 = 1.5 + 0.5 = 2$ h; $30/2 = 15$ km/h, which is the harmonic mean $2 \times 10 \times 30/(10 + 30) = 600/40 = 15$ because the legs are equal in distance. The plain mean, 20, is wrong. The **displacement** is zero (he is back at the start), so the average **velocity** is 0. (Verified.)
7. **Saturday.** 2099 is non-leap, so the one-year shift is one day. Before moving from 2100 to 2101, check the century rule: 2100 is not divisible by 400 and is therefore non-leap.
8. **The stated orientation is not feasible; rotated, 18 tiles.** $9/2$ is not integral, but on rotation $12/2 = 6$ and $9/3 = 3$, so $6 \times 3 = 18$. (Verified.)

9. Timed transfer and retry gate

Attempt the eight drills in **12 minutes**. For every error, label it C, A, X, R, or T using the table above. Retry only the missed family after rebuilding its diagram/equation. Move on only after at least **7/8**, with no unit, century-rule, or condition error; otherwise repeat the matching method and take a fresh equivalent set.

10. PYQ-pattern notes (2024-2026, Set A)

- ANALYSIS: This family is relatively stable in the audited sample: 7/80 in 2024, 6/80 in 2025, and a provisional 8/80 in 2026. Broad competence still beats specialising.
- ANALYSIS: Recurring shapes: **time-work / pipes, speed-distance and trains/boats**, ****clocks** (coincidence/angle), calendars (odd days), **and** cube-cutting / mensuration counts**.
- ANALYSIS: Two shapes worth rehearsing because the audited 2026 paper uses both: a ****segment-ordering** comparison (**two travellers, the same speeds, different segment order - Method 5**) and a forward/backward step item **where the net displacement****, not the path length, is asked.
- ANALYSIS: The papers reward the **rate-per-cycle** and **relative-speed** reductions; brute simulation is the time trap.

11. Study links

- FACT: Foundation companion.
- FACT: Arithmetic and Commercial Math - averages/ratios underpin average speed.
- FACT: Logical Reasoning, Coding, Counting and DI - cube counts overlap with combinatorial counting.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2018-2023.md. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2019, 2020, 2021, 2022, 2023
- **Paper(s):** CSAT
- **Routed question demands:** 45

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	5	Velocity-time distance ratio	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	6	Train crossing time	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	35	Painted-cube face geometry	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	40	Circular track speed ratio	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	58	Inscribed triangle count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	59	Calendar date sequence	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	60	Rectangle diagonal geometry	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	12	Calendar day determination	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	16	Race distance inference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	30	Right-triangle distance	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	35	LCM cycle coincidence	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	36	Race start scaling	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	49	Fast clock time correction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	57	Path displacement distance	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	59	Calendar year repeat	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	20	Calendar day calculation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	31	Compass direction tracking	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	38	Cube volume density ratio	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2020	CSAT	71	Combined work-rate fraction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	72	Multi-leg average speed	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	73	Circle-square intersection count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	74	Upstream-downstream speed ratio	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	80	Jump-sequence well climb	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	16	Bounce count sequence	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	18	Direction-distance path	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	20	Net displacement direction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	29	Two walkers meeting point	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	48	Calendar day calculation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	57	Clock-hand angle timing	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	63	Collinear-points ratio count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	68	Volume unit conversion	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	75	LCM calendar meeting day	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2021	CSAT	80	Work-rate remaining days	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	6	Direction distance tracking	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	7	Calendar day finding	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	15	Circular track overtaking	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	35	Time unit total conversion	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	46	Clock route time comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	57	Work rate comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	59	Rectangle cut feasibility	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	68	Clock hand coincidence	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	20	Cyclic work-rate schedule	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	49	Day-of-week calculation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	50	Periodic signal synchronisation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	67	Tile packing maximum	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- Velocity-time distance ratio
- Train crossing time

- Painted-cube face geometry
- Circular track speed ratio
- Inscribed triangle count
- Calendar date sequence
- Rectangle diagonal geometry
- Calendar day determination
- Race distance inference
- Right-triangle distance
- LCM cycle coincidence
- Race start scaling
- Fast clock time correction
- Path displacement distance
- Calendar year repeat
- Calendar day calculation
- Compass direction tracking
- Cube volume density ratio
- Combined work-rate fraction
- Multi-leg average speed
- Circle-square intersection count
- Upstream-downstream speed ratio
- Jump-sequence well climb
- Bounce count sequence
- Direction-distance path
- Net displacement direction
- Two walkers meeting point
- Clock-hand angle timing
- Collinear-points ratio count
- Volume unit conversion
- LCM calendar meeting day
- Work-rate remaining days
- Direction distance tracking
- Calendar day finding
- Circular track overtaking
- Time unit total conversion
- Clock route time comparison
- Work rate comparison
- Rectangle cut feasibility
- Clock hand coincidence
- Cyclic work-rate schedule
- Day-of-week calculation
- Periodic signal synchronisation
- Tile packing maximum

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

CONSOLIDATED REGISTER NOTES

Complete revision spine

- **Rate model:** Quantity equals rate multiplied by time.
- **Time and work:** Add work rates, not completion times.
- **Pipes and leaks:** Treat filling positive and emptying negative.
- **Speed and distance:** Match units before using distance equals speed times time.
- **Relative motion:** Add opposite-direction speeds and subtract same-direction speeds.
- **Trains and boats:** Include object length and current speed correctly.
- **Races and tracks:** Use relative distance on linear or circular paths.
- **Clocks:** Use the 5.5-degree-per-minute relative hand speed.
- **Calendars:** Reduce day shifts modulo seven and handle leap-year rules.
- **Geometry:** Use properties before coordinates or formulas.
- **Mensuration:** Distinguish length, area and volume units.
- **Verification:** Check dimensions, magnitude, endpoints and physical possibility.

Ownership and close-option firewall

- **Own here:** Owns time-work, pipes, speed-distance, relative speed, trains, boats, races, clocks, calendars, elementary geometry, mensuration and unit conversion.
- **Do not duplicate:** Commercial arithmetic belongs to Topic 03; algebraic comparison and sufficiency to Topic 05; spatial direction puzzles to Topic 06.
- **Verification:** Rate equations retain units. Geometry formulas are derived or decomposed, and generated solutions are checked dimensionally and by direct substitution.

Timed answer route

CLASSIFY -> EXTRACT -> REPRESENT -> EXECUTE -> VERIFY -> DECIDE

- Use estimation or option elimination only after preserving the governing condition.
- A blank costs zero; a rushed unsupported answer also consumes time and may attract negative marks.
- For every error, record concept/application/calculation/reading/passage/time/guess, repair the owner, and retry a new item rather than memorising the old option.